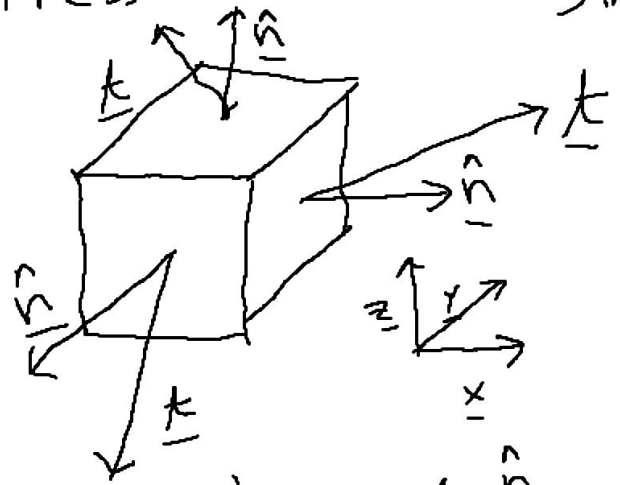
Stress Tensor

stress = force/unit area

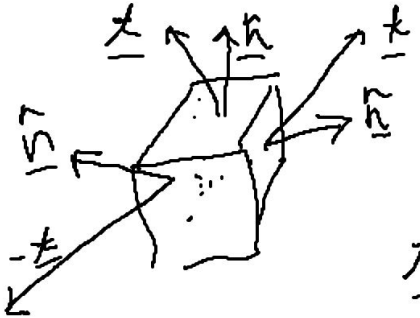


$$\underset{\text{stress tensor}}{\underline{\underline{\tau}}} \cdot \hat{\underline{n}} = \underset{\text{traction}}{\underline{t}}$$

normal stress $= \underline{t} \cdot \hat{\underline{n}}$

shear stress $= \underline{t} - \hat{\underline{n}}(\underline{t} \cdot \hat{\underline{n}})$ is force in plane $\perp \hat{\underline{n}}$

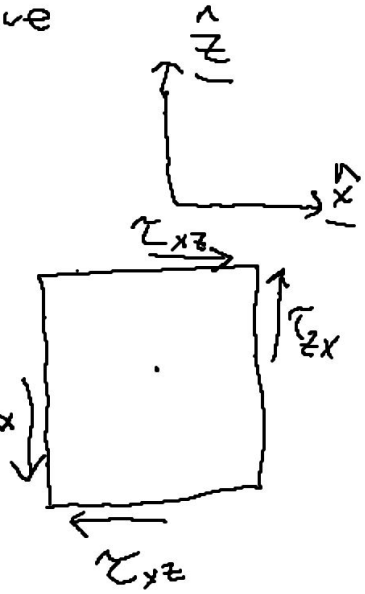
show that stress tensor is symmetric $\underline{\underline{\tau}} = \underline{\underline{\tau}}^T$



stress convention

extension is positive

$$\underline{t}(-\hat{\underline{n}}) = -\underline{t}(\hat{\underline{n}})$$

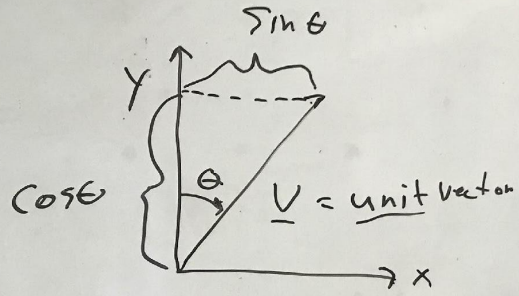


$$\underline{\underline{\tau}} = \begin{bmatrix} t_x(\hat{\underline{x}}) & t_x(\hat{\underline{y}}) & t_x(\hat{\underline{z}}) \\ t_y(\hat{\underline{x}}) & t_y(\hat{\underline{y}}) & t_y(\hat{\underline{z}}) \\ t_z(\hat{\underline{x}}) & t_z(\hat{\underline{y}}) & t_z(\hat{\underline{z}}) \end{bmatrix} = \begin{bmatrix} \tau_{11} & \tau_{12} & \tau_{13} \\ \tau_{21} & \tau_{22} & \tau_{23} \\ \tau_{31} & \tau_{32} & \tau_{33} \end{bmatrix}$$

to balance torques $\tau_{xz} = \tau_{zx}$

or $\tau_{ij} = \tau_{ji} \Rightarrow \underline{\underline{\tau}}$ is symetric. 6 independent elements.

Force/Area $=$ Pa $= N/m^2$ $\qquad N = kg\, m\, s^{-2}$



$\underline{\underline{\tau}}\,\hat{\underline{n}} = \lambda \hat{\underline{n}}$ prove

$\underline{\underline{e}}\,\hat{\underline{n}} = a\hat{\underline{n}}$ assume

azimuth = angle clockwise from North

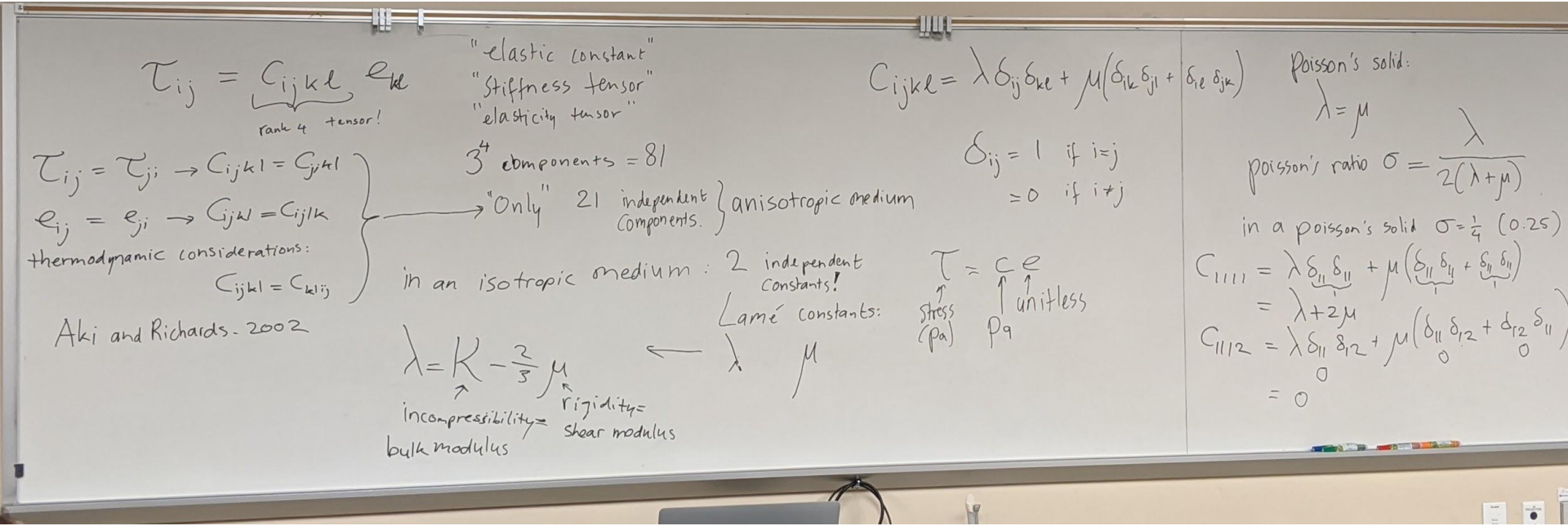
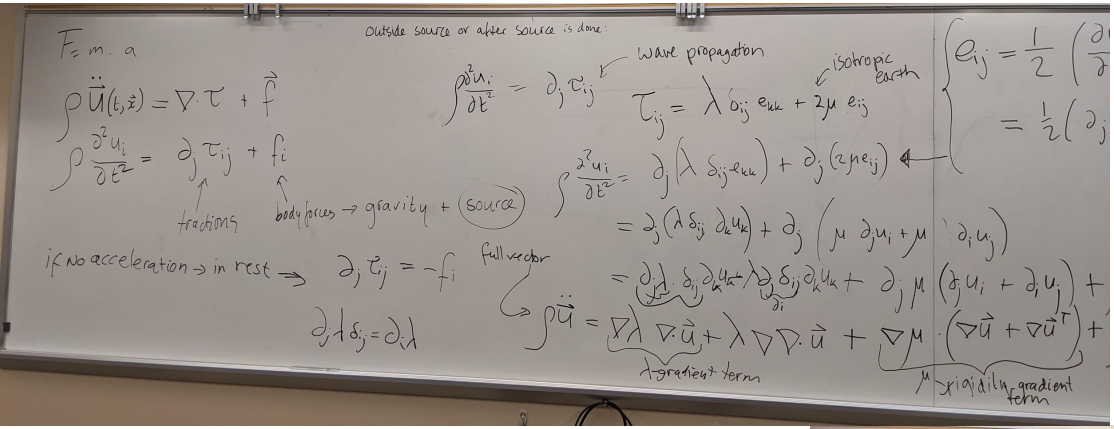
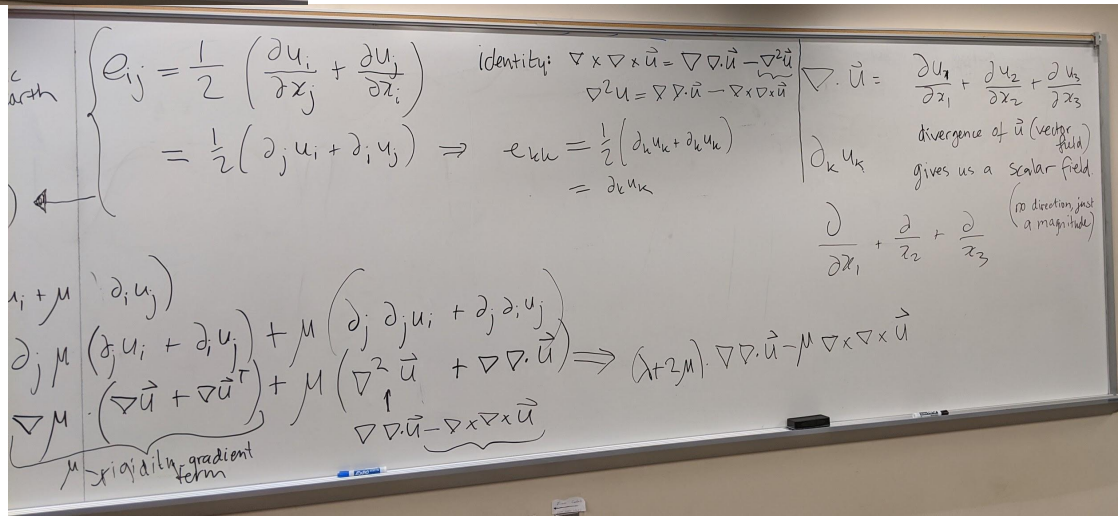
$$\boxed{\underline{\underline{\tau}} = \operatorname{tr}\underline{\underline{e}}\,\lambda \underline{\underline{I}} + 2\mu \underline{\underline{e}}}$$

$$\underline{V} = (\sin\theta, \cos\theta)$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{V_1}{V_2}$$

Deriving the seismic wave equation: elasticity tensor in anisotropic and isotropic medium

White board



Lecture 4 Wave equation continued